DC MOTOR POLARITY CONTROL

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Identify the main hardware components required to control a DC motor using an Arduino Uno and a Maker Drive motor driver.	☐
2	Safely connect a DC motor circuit using an external power supply and a common ground.	☐
3	Explain how an H-bridge changes the polarity across a DC motor to control clockwise and counterclockwise rotation.	☐
4	Use Simulink to manually control the direction and speed of a DC motor using digital signals and PWM values.	☐

2.0 Hardware Set-Up

This hardware set-up uses an Arduino Uno to control a DC motor through a Maker Drive motor driver, as shown in Figure 1. The Arduino sends control signals to the motor driver, while the motor driver supplies the higher current needed by the motor. A separate 9 V battery powers the motor side of the circuit, and the Arduino, motor driver and battery must share a common ground so that the control signals work correctly.

Required hardware

- ATmega328P microcontroller (Arduino Uno)
- USB Type-B cable
- Maker Drive motor driver board
- DC motor
- 9 V battery and clip
- Breadboard
- Breadboard jumper wires (various)

Hardware Assembly Steps

1. Connect the DC motor wires to one motor output channel on the Maker Drive, such as M1A and M1B.
2. Connect the positive wire of the 9 V battery to the motor power input of the Maker Drive (usually labelled VIN, VM or +).
3. Connect the negative wire of the 9 V battery to the GND terminal of the Maker Drive.
4. Connect the Maker Drive GND to one of the Arduino GND pins. This creates a common ground.
5. Connect the Arduino control pins to the Maker Drive input pins. In this exercise, Pin 9 drives input IN1 and Pin 10 drives input IN2.
6. Check all wiring carefully before powering the circuit. Make sure the motor is connected to the motor driver, not directly to the Arduino.
7. Connect the Arduino to the computer using the USB cable.

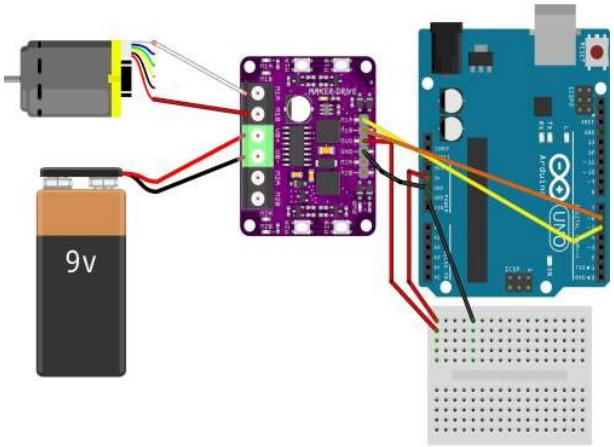


Figure 1: *Hardware Set-Up for DC Motor Polarity Control*

3.0 Simulink Set-Up and Results

The Simulink model for the H-bridge algorithm is shown in Figure 2, with the logic used to control the clockwise and counterclockwise motion of the DC motor. Manual Switch blocks are used here to set each input High or Low. To operate the motor on a physical system without the Simulink interface, the switches would need to be external, or the logic built into Simulink (e.g., using Saturation blocks, as described in later exercises). Based on the circuit above, the H-bridge inputs control the motor as shown in Table 2.

Table 2: *H-Bridge Input Logic for DC Motor Polarity Control*

Motion	IN1 (Pin 9)	IN2 (Pin 10)
Clockwise	High	Low
Counterclockwise	Low	High
Stop	Low	Low
Stop (alternative)	High	High

Once the model is set up and running in Simulink, carry out the following:

1. Change the Manual Switch positions as detailed in Table 2 and view the Scopes to see how the signals change.
2. The PWM blocks accept values between 0 and 255 (150 and 255 are used initially in Figure 2). Change the values in the Gain blocks to alter the speed of the DC motor.

To recap how to run Simulink code generation and deploy the model to the Arduino Uno, see Section 4.0 of the LED (Digital) Exercise or visit Code Generation.

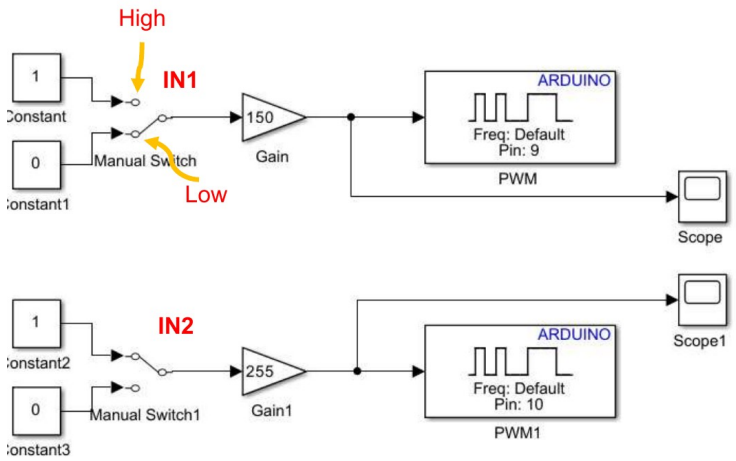


Figure 2: *Simulink Model for Manual Polarity Control of a DC Motor Using an H-Bridge*

4.0 Concluding Remarks

This exercise has demonstrated how an Arduino Uno and a Maker Drive motor driver can be used to control the direction and speed of a DC motor. By changing the H-bridge input signals, the polarity across the motor is reversed to produce clockwise or counterclockwise rotation, and PWM allows the motor speed to be adjusted. This provides a practical introduction to motor driver circuits, polarity control and real-time control implementation using Simulink, and a foundation for the following exercises on DC motor speed measurement and control.

Before completing the exercise, return to the Learning Outcomes in Table 1 and confirm that you are confident you can demonstrate each one. If so, tick the corresponding Achieved box.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

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